

Foundations of Computational Neuroscience - Project 4

Modelling a classical conditioning experiment for anticipatory shivering

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1 Project Description

In a classical conditioning paradigm, a rat is exposed to a tone that is followed by an ice-cold shower. The rat shivers when exposed to the cold water. Shivering is an instinctive response. After a number of trials, the rat starts to shiver when the tone is played. This is called anticipatory responding. The aim of this project is to develop an RL model for this experiment.

2 Model Proposal

The proposed model is based on the the papers [1], [2] and [3]. We model the rat's behavior as a Homeostatically Regulated Reinforcement Learning (HRRL) problem, in which actions are evaluated according to their effect on internal body temperature. The rat's internal state corresponds to its body temperature at time t , and the homeostatic setpoint corresponds to its normal body temperature T^* , which is set to 36 degrees in our simulations. The homeostatic drive is defined as the deviation from this setpoint:

$$d(t) = (T(t) - T^*)^2$$

The reward is then defined as the change in the homeostatic drive between two successive states:

$$R_t = d(t) - d(t+1)$$

This means that the closer an action brings the rat's body temperature to T^* , the more it is rewarded and vice versa; the more it increases the deviation from T^* , the less it is rewarded. In this model, shivering has a warming effect on the organism, meaning that it increases the rat's temperature by T_{shiver} degrees:

$$T(t+1) = T(t) + T_{\text{shiver}}, \quad \text{where } T_{\text{shiver}} > 0$$

while the cold shower has a cooling effect, meaning that it decreases the rat's temperature by T_{cold} degrees:

$$T(t+1) = T(t) - T_{\text{cold}}, \quad \text{where } T_{\text{cold}} > 0$$

Therefore, shivering is useful when the rat is cold, because it increases internal temperature and reduces the distance from the homeostatic setpoint. More importantly, if the rat learns that the bell predicts the cold shower, shivering can also become useful in the bell state as an anticipatory response: it raises body temperature before the cooling perturbation occurs, thereby limiting the future deviation from T^* .

3 State - Action Diagram

To formalize the classical conditioning task, we define three environmental states: an Initial state, a Bell state, and a Cold Shower state. In each of these states, the rat can choose between two actions: (i) no-shiver and (ii) shiver.

The diagram below presents the transition probabilities between the states, which are independent of the actions:

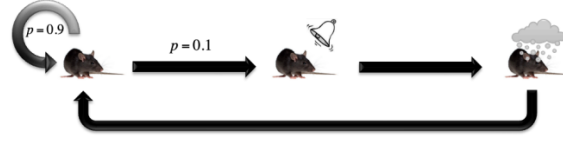


Figure 1: State Space

4 Plots

The code of our RL model can be found [here](#). The official plots, from paper [3] are the following:

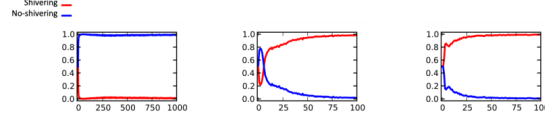


Figure 2: Original Plots from paper [3]

Our model results are presented below. Our plots match the results of the paper, proving that our model successfully captures the HRRL framework :)

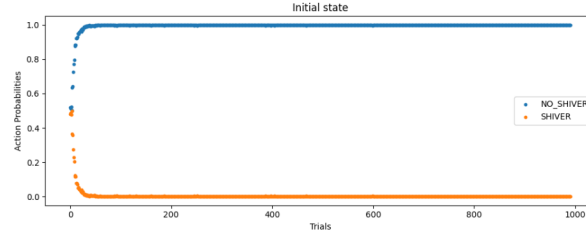


Figure 3: Learning Rate $\alpha = 0.2$, discounted factor $\gamma = 0.97$

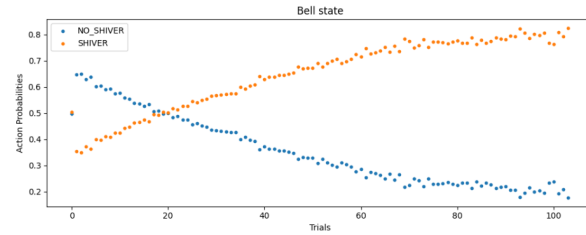


Figure 4: Learning Rate $\alpha = 0.2$, discounted factor $\gamma = 0.97$

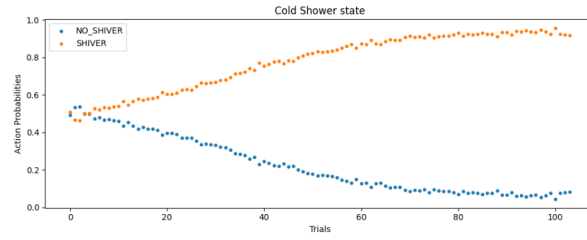


Figure 5: Learning Rate $\alpha = 0.2$, discounted factor $\gamma = 0.97$

References

- [1] Mehdi Keramati and Boris S. Gutkin. A reinforcement learning theory for homeostatic regulation. In *Advances in Neural Information Processing Systems 24 (NeurIPS 2011)*, pages 82–90. Curran Associates, Inc., 2011.
- [2] N. Keramati and B. Gutkin. Homeostatic reinforcement learning for integrating reward collection and physiological stability. *PLoS Computational Biology*, 10(11):e1003763, 2014.
- [3] Naoto Yoshida, Henning Sprekeler, and Boris Gutkin. Linking homeostasis to reinforcement learning: internal state control of motivated behavior. *Current Opinion in Behavioral Sciences*, 66:101611, 2025.